Pricing United States Treasury Securities

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LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

- **Classify Treasury securities.** Distinguish Treasury bills, notes, and bonds using their maturities and promised cash-flow structures.
- **Derive security prices.** Derive zero-coupon and coupon-bearing Treasury prices from the zero-NPV condition under a stated discounting convention.
- **Evaluate risk claims.** Assess which risks are suppressed—and which remain—when a Treasury security is treated as risk-free.

1 Why Does the Treasury Borrow?

When federal expenditures exceed tax and other receipts, the U.S. Department of the Treasury finances the resulting deficit by issuing debt. A Treasury security is therefore a contract: an investor supplies dollars today, and the federal government promises a specified pattern of future payments. The phrase *fixed income* describes those contractual payments; it does not imply that the security's market price remains fixed.

Marketable Treasury securities differ primarily in maturity and cash-flow pattern (Table 1). Bills promise one maturity payment, whereas notes and bonds promise periodic coupons plus repayment of principal. Those structural differences determine the appropriate abstract-asset model before any price is calculated.

Table 1: Current standard features of marketable nominal Treasury bills, notes, and bonds. The term is the time from issuance to maturity; market prices may change before maturity.

Security	Standard term	Promised cash flows	Interest timing
Treasury bill	4–52 weeks	Face value at maturity	At maturity
Treasury note	2, 3, 5, 7, or 10 years	Semiannual coupons and face value	Every six months
Treasury bond	20 or 30 years	Semiannual coupons and face value	Every six months

Treasury auctions translate bids into an issue rate, yield, or discount margin. Noncompetitive bidders accept the auction result, while competitive bidders specify an acceptable rate or yield. The auction allocates securities from the lowest acceptable financing cost upward, and all successful bidders receive the same accepted rate, yield, or margin. Valuation begins after those contract terms and a market discounting convention are stated.

2 Cash-Flow Structure Comes First

From the investor's perspective, purchasing any Treasury security creates a payment today and receipts later. The bill and coupon-security timelines expose the essential modeling difference: one future receipt versus a stream of coupons followed by principal repayment (Fig. 1). In both cases, arrows leaving a time-point unit denote money paid and arrows entering one denote money received.

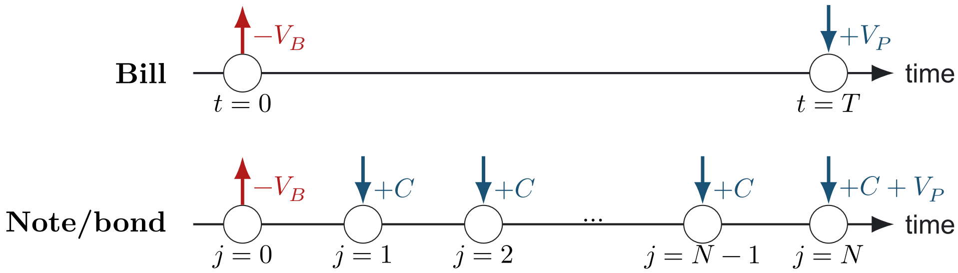

Figure 1: Cash flows from the investor's perspective. The purchase price V_B is paid at the valuation date. A bill returns only its face value V_P at maturity time T. A note or bond pays coupon amount C at each coupon index $j = 1, \dots, N$ and adds the face value to the final coupon at index N.

The one-payment structure in Fig. 1 defines a zero-coupon Treasury bill (Defn. 2.1) and separates its purchase price from its maturity value.

Definition 2.1. Treasury bill

Let time 0 be the purchase date and let the maturity time be $T > 0$ years. A Treasury bill is a zero-coupon debt security with purchase price V_B dollars at time 0 and face value V_P dollars paid at time T. The term *zero coupon* means that no contractual interest payment occurs between purchase and maturity. The nominal dollar gain at maturity is $V_P - V_B$; it is not itself a return. The continuously compounded holding-period growth rate and its associated dimensionless log return are

$$g_B := \frac{1}{T} \log\left(\frac{V_P}{V_B}\right), \qquad r_B := T g_B = \log\left(\frac{V_P}{V_B}\right). \tag{1}$$

A quoted Treasury yield follows its own market convention and should not be identified with g_B without the appropriate compounding and day-count conversion.

The absence of intermediate cash flows makes a bill the cleanest application of L1b's rule *convert, then add*. Under a specified model, its purchase price is the present value of the single promised receipt (Thm. 2.1).

Theorem 2.1. Zero-coupon price under periodic compounding

Let the positive integer n be the number of compounding intervals per year, let the maturity time be $T > 0$ years, and let the constant nominal annual yield used for valuation be y with $1 + y/n > 0$. Define its continuously compounded equivalent (with units of inverse years) by $g_y := n \log(1 + y/n)$ and the forward accumulation factor by

$$\mathcal{D}_{T,0}(y; n) := \left(1 + \frac{y}{n}\right)^{nT} = e^{g_y T}. \tag{2}$$

The real exponent nT need not be an integer for a zero-coupon claim. If the bill's face value V_P is paid with certainty at maturity and taxes, transaction costs, and other market frictions are ignored, then the zero-NPV purchase price V_B is given by Eq. 3:

$$V_B = \mathcal{D}_{T,0}^{-1}(y; n) V_P = \left(1 + \frac{y}{n}\right)^{-nT} V_P = e^{-g_y T} V_P. \tag{3}$$

Here $\mathcal{D}_{T,0}^{-1}(y; n) := 1/\mathcal{D}_{T,0}(y; n)$ converts maturity-date dollars to time-0 dollars. Under this flat periodic-yield model, Eq. 3 implies $g_B = g_y$; market quotation conventions may imply a different conversion.

The bill-pricing relation in Eq. 3 follows by setting the investor's NPV equal to zero and solving for the only time-0 cash flow:

$$\begin{aligned} 0 &= -V_B + \mathcal{D}^{-1}_{T,0}(y;n)V_P, && \text{zero NPV at the purchase price,} \\ V_B &= \mathcal{D}^{-1}_{T,0}(y;n)V_P, && \text{isolate the time-0 price.} \end{aligned}$$

The compounding frequency n and nominal yield y in this course model must be stated explicitly. Treasury bill auction quotations also use market-specific bank-discount and investment-rate conventions; those quotation rules should not be silently identified with an arbitrary compound-interest model or with the continuously compounded growth rate g_y.

The companion notebook applies Thm. 2.1 to auction records and makes the quotation convention visible in the calculation.

> **COMPANION NOTEBOOK**
> **L2a: Treasury Bill Pricing Worked Example →**
> Compute zero-coupon bill prices from auction inputs and compare model values with reported prices.

3 Coupon-Bearing Notes and Bonds

Treasury notes and bonds spread value across time rather than concentrating it at maturity. That stream creates both an income component and greater sensitivity to how discount rates vary across maturities.

The repeated-payment structure in Fig. 1 defines a coupon-bearing Treasury security (Defn. 3.1) and distinguishes coupon income from principal repayment.

> **Definition 3.1. Coupon-bearing Treasury security**
>
> Let the positive integer n be the number of coupon payments per year, let the maturity time be $T > 0$ years with $N = nT$ coupon dates, let the face value be V_P dollars, and let the annual coupon rate be c. A nominal coupon-bearing Treasury security pays the coupon amount $C := (c/n)V_P$ dollars at every coupon date $j = 1, \dots, N$ and repays the face value V_P with the final coupon. Treasury notes and bonds share this cash-flow structure but have different standard maturity ranges.

Each promised receipt must be converted separately because it belongs to a different dated currency. Under a flat nominal yield, adding those time-0 values produces the coupon-security pricing relation (Thm. 3.1).

Theorem 3.1. Coupon-security price under a flat yield

Let the positive integers n and $N = nT$ denote the coupon frequency per year and total number of coupons over the maturity time T years. Let V_P be the face value, let C be the coupon amount, and let the constant nominal annual yield used for valuation be y with $1 + y/n > 0$. Define the accumulation factor to coupon date j by $\mathcal{D}_{j,0}(y;n) := (1 + y/n)^j$. If all promised payments are made with certainty and taxes, transaction costs, accrued-interest conventions, and other market frictions are ignored, then the zero-NPV purchase price is given by Eq. 4:

$$V_B = \sum_{j=1}^{N} \mathcal{D}_{j,0}^{-1}(y;n)C + \mathcal{D}_{N,0}^{-1}(y;n)V_P. \tag{4}$$

The summation index j enumerates coupon dates, and each reciprocal factor converts that date's dollars to time-0 dollars.

The two terms in Eq. 4 separate the economic claims: the sum is the present value of the coupon stream, and the final term is the present value of principal repayment. The zero-NPV derivation is therefore:

$$\begin{aligned}
0 &= -V_B + \sum_{j=1}^{N} \mathcal{D}_{j,0}^{-1}(y;n)C + \mathcal{D}_{N,0}^{-1}(y;n)V_P, && \text{value every signed cash flow at time 0,}\\
V_B &= \sum_{j=1}^{N} \mathcal{D}_{j,0}^{-1}(y;n)C + \mathcal{D}_{N,0}^{-1}(y;n)V_P, && \text{solve for the purchase price.}
\end{aligned}$$

The note-and-bond notebook evaluates this cash-flow sum using actual auction data; it is placed here because its code implements the coupon-bearing model directly.

COMPANION NOTEBOOK

L2a: Treasury Note and Bond Pricing Worked Example →

Construct coupon cash flows, discount each payment, and compare the resulting value with auction prices.

4 What Does "Risk-Free" Suppress?

Calling a Treasury security risk-free is a modeling statement, not a description of every possible outcome. In a one-period valuation model, the phrase usually means that the promised dollar payment is treated as known and default is excluded. That assumption makes Treasury yields useful benchmark rates, but it does not conserve purchasing power or market value.

Credit and political risk. The probability that the U.S. Treasury fails to make a promised nominal payment is generally treated as very low, but institutional and political disruptions are not logical impossibilities.

Interest-rate and liquidity risk. A holder who sells before maturity receives the prevailing market price, not the contractual face value. When market discount rates rise, the present value of fixed promised payments falls; L2b develops this sensitivity precisely. Treasury markets are highly liquid, but bid-ask spreads and market depth can still affect an immediate sale price.

Inflation and reinvestment risk. Nominal repayment does not guarantee purchasing power. Coupon-bearing securities also require the investor to reinvest intermediate coupons at rates that are unknown when the security is purchased.

The useful engineering habit is to attach the qualifier to the claim: risk-free *with respect to which uncertainty, over what holding period, and under which model assumptions?*

KEY TAKEAWAYS

In this lecture, we represented Treasury bills, notes, and bonds as abstract assets and derived their prices by discounting promised cash flows to a common valuation date.

- **Structure determines value.** Bills contain one maturity receipt, while notes and bonds contain coupon streams plus principal; the cash-flow structure must be identified before pricing.
- **Price means zero NPV.** Under stated assumptions, a Treasury's purchase price is the amount that makes the investor's discounted signed cash flows sum to zero.
- **Risk-free is conditional.** Treating nominal payments as certain suppresses default risk but does not eliminate interest-rate, inflation, reinvestment, or liquidity concerns.

Next, use these pricing relations to determine how a bond's value changes when its yield, coupon, maturity, or payment timing changes.

ADDITIONAL READING

- U.S. Treasury, *Treasury Bills*, for current terms, cash-flow conventions, and auction links.
- U.S. Treasury, *Treasury Notes* and *Treasury Bonds*, for current maturities and coupon-payment conventions.
- U.S. Treasury, *How Auctions Work*, for competitive and noncompetitive bidding mechanics.